

A Sparse 3D GAN-Based Mixture-of-Experts Framework for Object Generation and Completion



Yahia Hamdi¹, Nicolas Andrialovanirina¹, Kelig Mahe², Emilie Poisson-Caillault¹

¹ Université du Littoral Côte d'Opale, UR 4491, LISIC, F-62100 Calais

² Ifremer, Unité HMMN, 150 quai Gambetta, F-62321 Boulogne-sur-Mer

mail:emilie.poisson@univ-littoral.fr

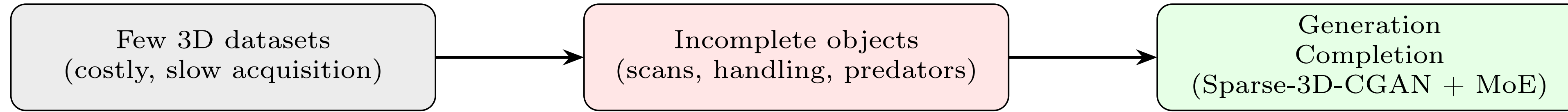


The Challenge: Scarce & Incomplete 3D Data

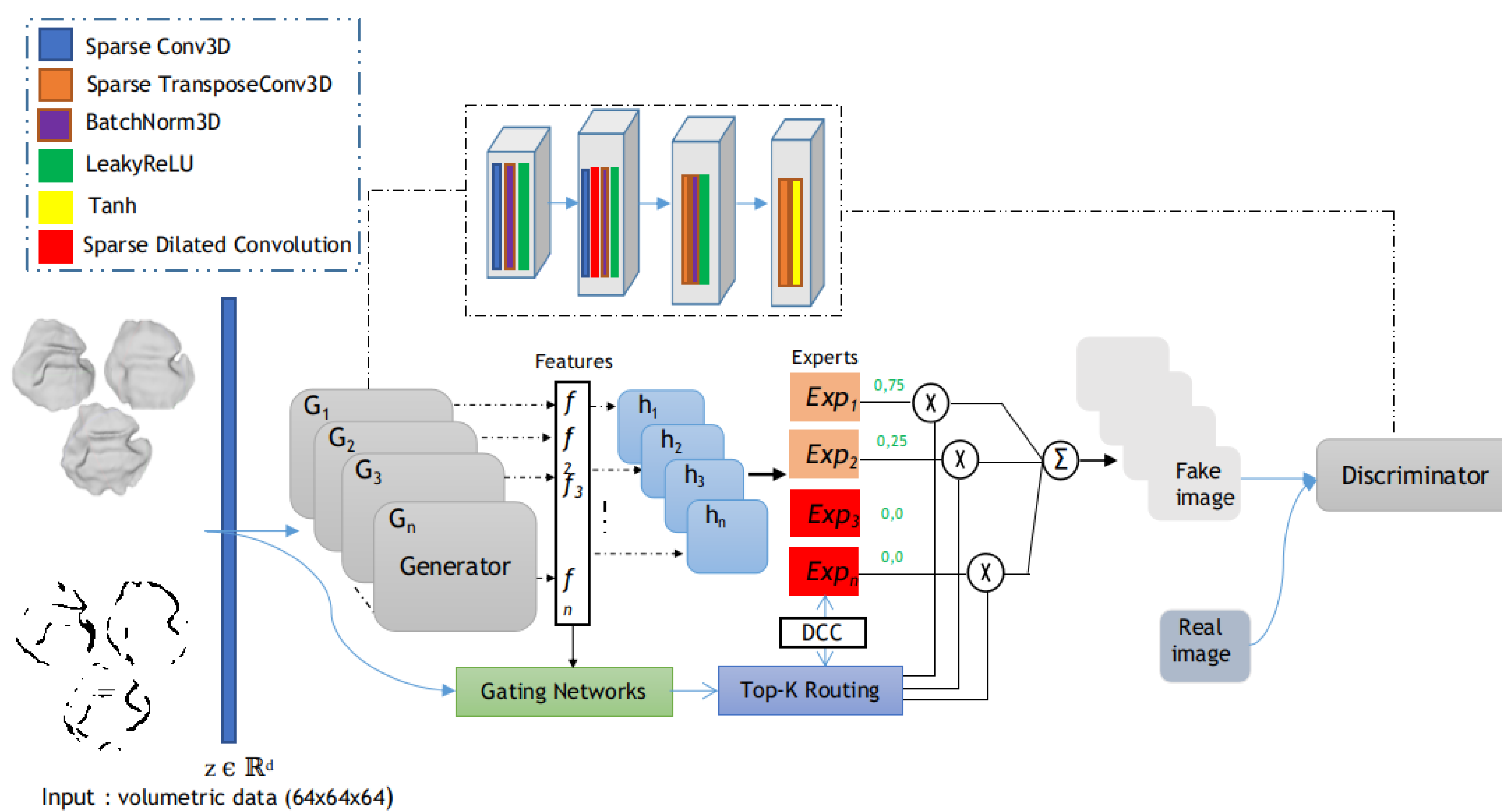
High-quality 3D datasets are scarce in domains like marine biology, where data collection is costly, ethically constrained, or limited to **damaged/incomplete objects**.

This is critical for **otolith** analysis: their complex morphology encodes species identity and environmental conditions, essential for classification and fisheries management.

⇒ Need methods that **generate** plausible shapes *and* **complete** sparse, damaged, occluded ones



Method: Sparse-3D-CGAN with MoE

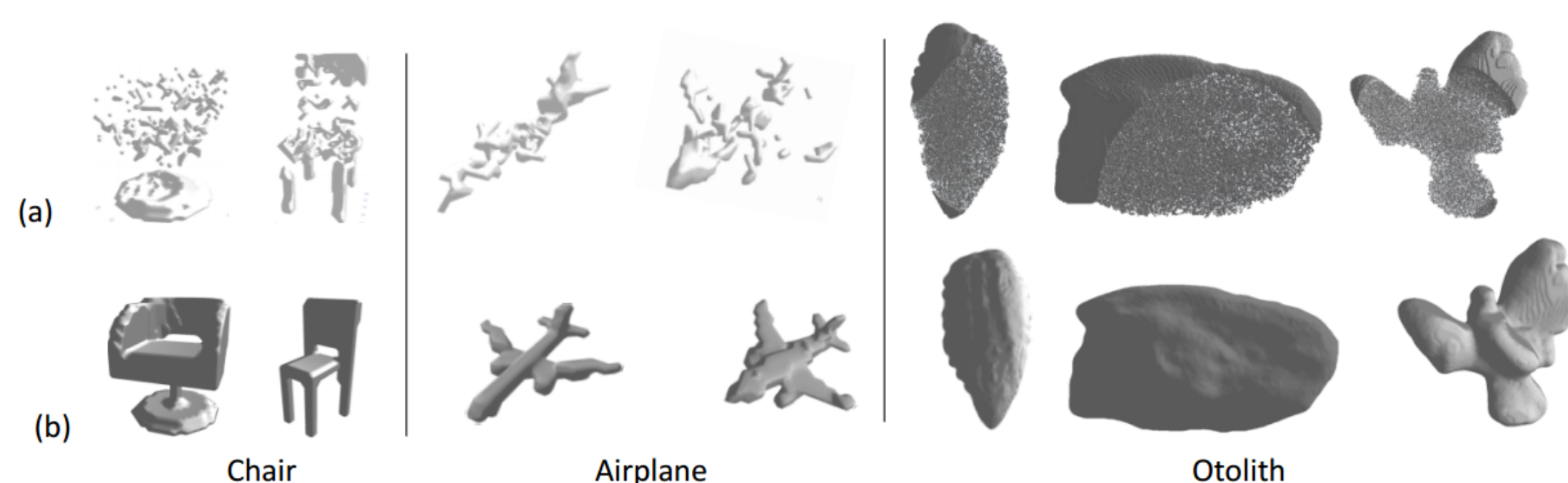


Key components:

- n **sparse-conv expert generators**, each specializing in distinct geometric patterns via encoder-decoder with SparseConv3d/SparseConvTranspose3d
- **Shared discriminator** across all experts for adversarial training
- **Context-aware gating**: $g_i = p_i$ if $i \in \text{top}_k(\mathbf{p})$, else 0
- **DCC - Dynamic Capacity Constraint**: capacity $C_i^{(t+1)} = C_b(1 + \alpha(\frac{w_i}{n} - U_{\text{avg}}[i]))$ balances experts *without* auxiliary loss

3D objet completion

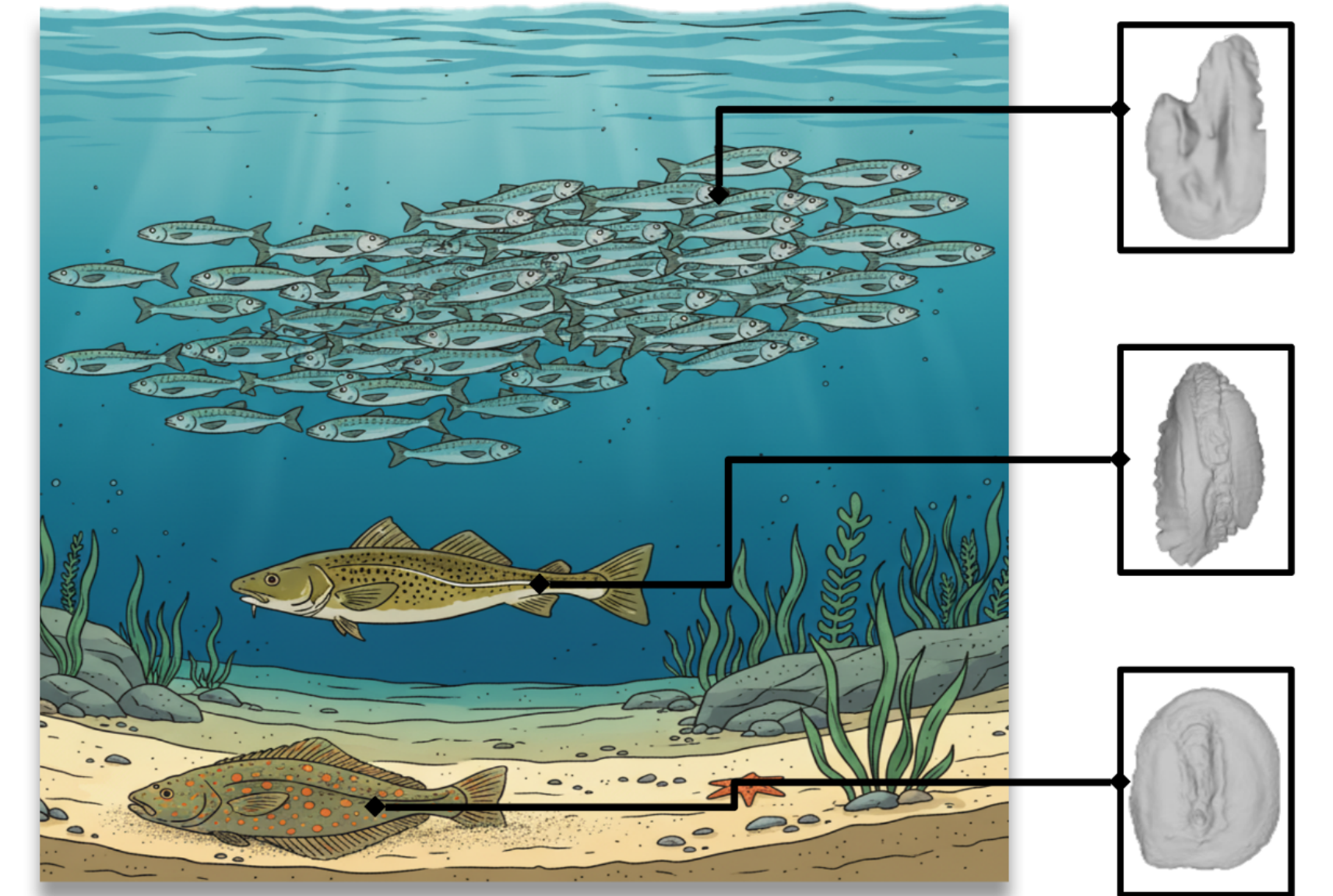
Comparisons of completion results using sparse MoE-CGAN on Chair, Airplane, and Otolith shapes.
(a) Partial object, (b) Reconstructed object.



References

- [1] 3D Fish Otolith data: 10.17882/106566 and 10.17882/106566
- [2] ShapeNet data: <https://doi.org/10.48550/arXiv.1512.03012>
- [3] 3D-GAN, in Proc. NeurIPS 2016, doi.org/10.48550/arXiv.1610.07584
- [4] RAGans, in Proc. ICLR'2019, doi.org/10.48550/arXiv.1807.00734
- [5] DiffComplete, in Proc. NeurIPS 2023, doi.org/10.48550/arXiv.2306.16329
- [6] PrGAN, in *Int Journal of Computer Vision 2019*, doi.org/10.48550/arXiv.1906.04910
- [7] GAN, in *Proc. Eurographics*, 2020. doi.org/10.48550/arXiv.2002.00349

Application Context



Otolith = the fish's "black box" ear bone
From Otolith many fish traits could be identified :
species, habitat, fish fins, maturity age.

⇒ **Better fisheries stock management**

Protocol and Results

- 3 datasets : Chair and Airplane from ShapeNet benchmark [1] and Otolith [2].
- 70% of original points masked .
- Train/Test split ratio : 80%-20%
- **4 metrics**: 3 distance-based metrics measuring shape similarity **Chamfer Distance (CD)**, **Hausdorff Distance (HD)**, and **Earth Mover's Distance (EMD)**, and one indicator of fidelity structure **Percentage of Points Retained Rate (PRR)** from the input shape.

Table: Completion score on Chair, Airplane and Otolith datasets. CD ($\times 10^2$), HD ($\times 10^3$), EMD ($\times 10$), PRR.

Cat.	Model	CD	HD	EMD	PRR
Airplane	3DGAN [3]	46.7	72.8	84.0	77.2
	RaGANs [4]	41.2	71.3	81.1	85.7
	GAN [7]	16.9	65.1	63.1	-
	PrGAN [6]	15.1	64.2	60.3	-
	DiffComplete [5]	12.7	-	55.9	-
	MoE-CGAN (ours)	14.5	62.1	60.4	95.4
Chair	3DGAN [3]	43.2	76.0	80.4	82.5
	RaGANs [4]	41.1	73.2	79.6	83.4
	GAN [7]	34.2	65.1	63.1	-
	PrGAN [6]	26.1	64.2	60.3	-
	DiffComplete [5]	20.3	-	55.1	-
	MoE-CGAN (ours)	25.4	63.8	59.5	95.5
Otolith	3DGAN [3]	54.8	94.1	81.9	80.0
	RaGANs [4]	52.6	87.6	79.1	82.3
	MoE-CGAN (ours)	40.9	80.2	68.5	94.6

Contributions

- A frugal **Sparse-3D-CGAN + MoE** architecture tackling mode collapse and instability for otolith completion/generation
- $3\times$ fewer FLOPs and $7\times$ faster inference than diffusion models
- relevant with context-aware routing.
- **Geometry-aware specialization**: multiple 3D CGAN experts, balanced via auxiliary-loss-free DCC
- **Shape fidelity and morphological diversity** across multiple datasets

Scan — quite fun!

